JAN–S–726

27 JANUARY 1949

NATIONAL MILITARY ESTABLISHMENT SPECIFICATION

SHELLS, SHOTGUN, 12-GAGE

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

U. S. ARMY
50–5–1C
10 Feb. 1944

A. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS, AND DRAWINGS

A–1. *Specifications.*—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

U. S. ARMY SPECIFICATIONS

52–18–1—Shotguns, 12–Gage, Sporting and Riot Type.
100–2—Standard Specification for Marking Shipments by Contractors.
57–0–2—Finishes, Protective, for Iron and Steel Parts.

A–2. *Other publications.*—The following publications, of the issue in effect on date of invitation for bids, form a part of this specification:

Navy Shipment Marking Handbook.
Technical Committee Manual of the Sporting Arms and Ammunition Manufacturers' Institute.

A–3. *Drawings.*—The following drawings, of the issue in effect on date of invitation for bids, shall form a part of this specification:

U. S. ARMY ORDNANCE DEPARTMENT DRAWINGS

D7640981—Shell, Shotgun, All-Brass, No. 00 Buck, M19.
C7673530—Packing and Marking for Shell, Shotgun, All-Brass, 12-Gage, No. 00-Buck, M19.
C7673531—Packing and Marking for Shell, Shotgun, 12-Gage.
D43655B—Receiver, Universal, M1, Assembly.
D43621B—Receiver, Universal, M2, Assembly.

B. TYPES AND CLASSES

B–1. This specification covers the following types and classes of shells, shotgun, 12-gage:

Type I.—All Brass, No. 00 (Eastern) Buckshot, M19, for Guard and Combat.
Type II.—Paper Cases with "Medium" (0.73 inch min.) Metal Heads, No. 00 (Eastern) Buckshot for Guard or Combat.
Type III.—Paper Cases with "low" (0.35 inch min.) Metal Heads (Standard Field Grade).
Class A—Trap No. 7½ or No. 8 Standard Shot.
Class B—Skeet, No. 9 Standard Shot.
Class C—Hunting, as specified in contract.

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C. MATERIAL AND WORKMANSHIP

C–1. The brass used in making shell, shotgun, 12-gage, all-brass, No. 00 buckshot, M19, shall be standard cartridge brass.

C–2. The metal used in the head of the paper case shall be standard cartridge metal, steel or brass, as specified on drawing or in contract.

C–3. When either the shell head or a primer cup (or primer battery cup), or both, is or are made of steel, it or they shall be given a corrosion-resistant finish conforming to commercial standards.

C–4. The paper part of the case shall be of first-class shell case body paper processed according to standard commercial practice.

C–5. The shell, primer, powder, shot, and wadding shall be manufactured in accordance with the best commercial practice.

D. GENERAL REQUIREMENTS

D–1. The initials or other identifying marks of the contractor shall be stamped on the metal head of the shell.

D–2. *Primer.*—The primer shall be a standard noncorrosive primer of commercial manufacture suitable for proper ignition of the charge of propellent powder used. In type I and type II shells, after the battery cup and primer are seated in the case, the circular joints around the primer and battery cup shall be waterproofed with material colored to indicate its presence.

D–3. *Wadding.*—The wadding shall be of the size, type, quality, and quantity regularly used in standard commercial practice for the particular shell and load specified in the contract. Wadding shall not be tipped, base-blown, or disturbed to the extent that the load is exposed.

D–4. *Powder.*—The propellent powder shall be any approved clean burning smokeless powder normally used in commercial loading for the type of shell and load specified.

D–5. *Shot.*—The size of the buckshot required under this specification shall be "Eastern," and other shot mentioned herein shall be "American Standard," with the number of pellets and tolerance to the ounce as stated in the Technical Committee Manual of the Sporting Arms and Ammunition Manufacturers' Institute, unless otherwise specified.

D–6. The shell shall be center-fire; and type II and type III shells shall be fitted with base wads suitable for the type and quantity of powder used.

D–7. The assembled shell, before crimping, shall have an over-all length according to standard commercial practice for the particular shell and load specified in the contract; and the crimped assembly shall be snug and without voids among the internal components, other than the unavoidable interstices among the individual powder grains and shot pellets.

D–8. *Visual defects and chambering.*—

D–8a. The shells shall be inspected visually for the defects listed in table I.

D–8b. The shells shall be placed in the magazine to its capacity and shall then be chambered from the magazine by the action of the gun, but not fired.

D–8c. Defects discovered by visual inspection, chambering, function and casualty, and all tests shall not exceed the percentages listed in table I.

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D–9. *Rustproof.*—When shot shells are made according to paragraph C–3, such heads or cups shall show no sign of rust after being subjected to the test specified in paragraph F–7.

TABLE I.—*Defects and limits permitted in a sample.*

Defects [1]	Percent	
	Types I and II	Type III
Visual defects which may cause malfunctions [2]	3.0	2.0
Failure to magazine	.33	.66
Failure to chamber	.33	.66
Failure to extractor eject	.33	.66
Head pulled off	0	.1
Heat split, through head	0	.1
Split mouth	2.0	2.5
Head start less than 1/16 inch	2.0	2.5
Head start more than 1/16 inch	.33	.66
Rupture	.33	.33
Head cut off	0	0
Partial cut off	.33	.33
Body cut off	0	0
Bulged rim	2.5	2.5
Partial split, less than shell length	.33	.66
Body split, full length	0	0
Burst rim	0	0
Split knurl	2.0	2.5
Powder burns	2.0	2.5
Blown battery or primer cup	0	.2
Blown base wad	0	.2
Primer blown up	0	.2
Primer set back	.33	.66
Primer leak	1.0	2.0
Primer punctured	2.0	2.5
Primer pierced	.5	1.0
Misfire	.33	.66
Tipped wads	0	.2
Crimped opening, exposing load [3]	0	----------

[1] See chart I for illustrations of terms used when reporting shot shell defects.

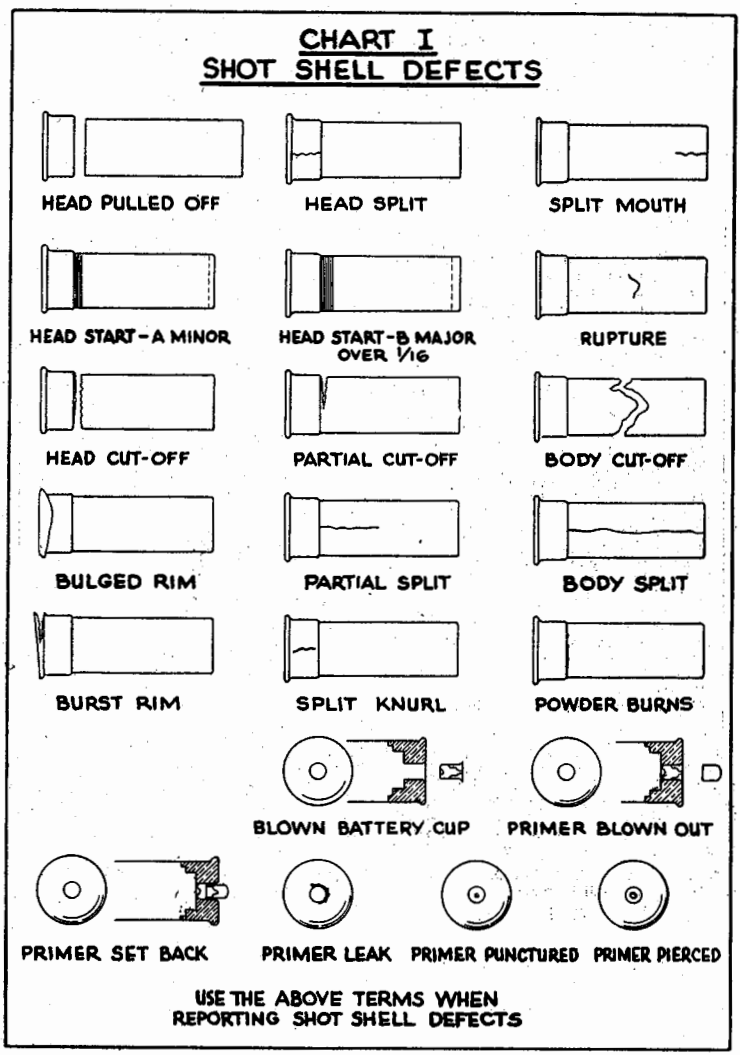
[2] Shells having visual defects, which, in the belief of the Government inspector will cause malfunctions, shall be segregated at the time of visual inspection and tested separately for chambering through the magazine by the action of the gun, and fired. Such shells which chamber through the magazine and fire satisfactorily shall not be counted as defects. This does not apply to shells which have been subjected to the water absorption test.

[3] Applicable to type II only.

NOTE.—If any calculation above results in a fractional number of defects, the next higher whole number shall be used.

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E. DETAIL REQUIREMENTS

E–1. *Water absorption.*—Type II shells which have been submerged in water shall feed in the magazine to its capacity and shall then be chambered from the magazine by the action of the gun, but not fired.

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Chambering and magazining failures shall not exceed the percentages shown in table I.

E–2. *Velocity.*—

E–2a. The average velocity at 60 feet from the muzzle disjunctor and the shot charge for each shell named below are:

Type I—1125 plus or minus 20 feet per second, with 9 buckshot No. 00.

Type II—1125 plus or minus 20 feet per second, with 9 buckshot No. 00.

Type III—Class A—870 plus or minus 20 feet per second, with 1⅛ ounces No. 7½ shot; and 855 plus or minus 20 feet per second, with 1⅛ ounces No. 8 shot. Class C—Velocity and shot charge as specified in the contract.

E–2b. The average velocity at 37½ feet from the muzzle disjunctor and shot charge for the shell named below is:

Type III—Class B—920 plus or minus 20 feet per second, with 1⅛ ounces of No. 9 shot.

E–2c. A correction factor of one foot per second per degree Fahrenheit shall be applied for correcting to a normal range temperature of 70° F. when velocities are recorded over 25 and 40 yard ranges. When the range temperature is below 70° F., correction shall be added. When the range temperature is above 70° F., correction shall be subtracted. Range temperature shall be recorded in the range, but not in the firing room.

E–3. *Pressure.*—The maximum average pressure for each type of shell shall not exceed 11,000 pounds per square inch.

E–4. *Pattern.*—

E–4a. The following shall be the minimum average patterns in a 30-inch circle at 120 feet plus or minus one foot from the gun muzzle for the types and classes named:

Type I—66.7 percent, based on actual average pellet count from shells taken from the test sample.

Type II—66.7 percent, based on actual average pellet count from shells taken from the test sample.

Type III—Class A—60 percent, based on pellet count of 394 plus or minus 10 for No. 7½ standard shot, with pellet diameters of 0.095 plus or minus 0.0025 inch; 60 percent, based on pellet count of 460 plus or minus 12 for No. 8 standard shot, with pellet diameters of 0.090 plus or minus 0.004 inch. Class C—As specified in the contract.

E–4b. The following shall be the minimum average pattern in a 30-inch circle at 75 feet plus or minus one foot from the gun muzzle for the type named:

Type III—Class B—50 percent, based on an actual average pellet count from five shells taken from the test sample.

E–4c. A correction factor of one percent for each 5° F. temperature variation shall be applied for correcting to a normal range temperature of 70° F. when the variation is more than plus or minus 15° from 70° F. at the time of firing pattern tests. When the temperature is below

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70° F., correction shall be added. When the temperature is above 70° F., correction shall be substracted. Range temperature shall be recorded in the range, not in the firing room.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Place of inspection.*—Inspection and test for acceptance of the finished shot shells, unless otherwise specified, shall be made at Frankford Arsenal.

F–2. *Lots.*—The standard size lot shall be not more than 1,000,000 nor less than 50,000 rounds. However, when requested by the contractor and approved by the Government inspector, the lot may contain less than 50,000 rounds. The size of a lot may be increased by inclusion of the cartridges of the sample which are returned to the contractor, as provided for in paragraph F–3. Each lot shall consist of cartridges fabricated by the same processes from like materials and identified by lot number furnished by the contracting officer. The test sample shall not be considered as deliveries under the contract.

F–3. *Sampling.*—The Government inspector shall select 1500 rounds representative of each lot for the purpose of inspection and tests. Shell cartons shall not be opened except to obtain the cartridges needed for tests required by table II. The inspection sample shall be marked by the inspector so that the unopened cartons can be identified with the lot number. That portion of the sample which has not been required for the test or retest shall be returned to the contractor, provided no cartridges have been removed from the carton. All sample shells other than those in unopened cartons remaining after completion of the test shall be destroyed. The test sample shall not be considered as deliveries under the contract.

TABLE II.—*Acceptance samples.*

[Size of lot not over 1,000,000]

Test	Number of rounds
Chambering	100
Water absorption	20
Rustproof	20
Velocity and pressure	20
Pattern	20
Function and casualty	300
Visual inspection	500
Size of acceptance samples, all tests, including a possible retest	1,500

F–4. *Visual inspection.*—All shells in the opened cartons of the test sample shall be first inspected visually. Shells having visual defects which, in the belief of the Government inspector, will not function in the shotgun, shall be segregated and tested in accordance with note 2 to table I.

F–5. *Chambering test.*—

F–5a. For type I and type II shells, the chambering test shall be made by equally distributing the quantity of shells in the following shotguns:

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Manufacturer	Type	Model	Barrel length	Magazine capacity
			Inches	
Remington Arms Co., Inc.	Autoloading	11 (riot)	20	2 or 4
Winchester Repeating Arms Co.	Pump	97 (riot)	20	2 or 5

F–5b. For type III, classes A, B, and C, shells, the chambering test shall be made by equally distributing the quantity of shells in the following shotguns:

Manufacturer	Type	Model	Barrel length [1]	Magazine capacity
			Inches	
Ithaca Gun Co	Pump	Sporting	26, 28, 30	2 or 4
Savage Arms Co	Autoloading	720	26, 28, 30	2 or 4
Stevens Arms Co	Pump	620A	26, 28, 30	2 or 5
Remington Arms Co., Inc.	Pump	31	26, 28, 30	2 or 4
Winchester Repeating Arms Co.	Pump	12	26, 28, 30	2 or 5

[1] Any of the barrel lengths shown may be used.

F–5c. Shotguns used in tests shall conform to specification No. 52–18–1.

F–6. *Water absorption test (required for type II only).*—After both the shells and the water have been brought to room temperature, the shells shall be placed horizontally in the water under a 1-inch head over the rim, and so remain a continuous two hours. After the elapsed time, the shells shall be removed from the water, wiped dry, and magazined and chambered in accordance with paragraph D–8b in equal quantities in both of the shotguns mentioned in paragraph F–5a.

F–7. *Rustproofing test.*—

F–7a. *Salt spray.*—The test shall consist of the continuous 2-hour exposure of the heads and cups to the finely divided spray of a 20 percent (weight) sodium chloride solution at a temperature of not less than 95° F. plus 2° F. or minus 3° F. The test shall be conducted in a closed cabinet of material nonreactive to the salt spray. The cabinet shall be provided with an inclined cover so arranged as to prevent dripping of the condensed liquid on the specimen and so designed as to prevent any liquid which has once been in contact with the specimen from returning to the aspirator for respraying. The cabinet shall be so baffled that there will be no direct impinging of the spray upon the test specimens. If the spray solution is atomized by means of compressed air, the compressed air shall be clean and have a minimum relative humidity of 90 percent. (See specification No. 57–0–2 for description of salt spray test.

F–7b. The following test may be substituted for the salt spray test: The cartridges shall be stored in an atmosphere having a temperature of from 115° F. to 120° F. and a relative humidity of from 75 to 80 percent for a period of one week, after which they shall be exposed

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to ordinary room temperature and humidity for at least two hours before examining for signs of rust.

F–8. *Velocity and pressure test.*—

F–8a. The velocity and pressure for type I and type II and for type III, classes A and C, shall be determined simultaneously in a combination velocity and pressure barrel having a full choke 30-inch barrel with minimum to mean chamber and with chamber headspace and barrel dimensions conforming to the specifications of the Sporting Arms and Ammunition Manufacturers' Institute, as shown in its Technical Committee Manual. The barrel shall be assembled to a universal receiver, M1 or M2. The velocity shall be determined over a range of 120 feet from muzzle disjunctor to terminal target with the muzzle disjunctor not more than one foot from gun muzzle. The terminal target shall be the standard 5/16-inch shot shell target plate, approximately 4 by 4 feet.

F–8b. The velocity and pressure for type III, class B, shall be determined simultaneously in a combination velocity and pressure barrel having an improved cylinder or "skeet," bored 26-inch barrel with chamber, headspace, and barrel dimensions conforming to Sporting Arms and Ammunition Manufacturers' Institute specifications, as described in paragraph F–8a. The velocity shall be determined over a range of 75 feet from muzzle disjunctor to terminal target with the muzzle disjunctor not more than one foot from the gun muzzle. The terminal target shall be the standard described in paragraph F–8a.

F–8c. Reference trap load and skeet load ammunition loaded respectively with 1⅛ ounces number 7½ standard shot to an instrumental velocity at 60 feet of 870 plus or minus 20 feet per second and with 1⅛ ounces number 9 standard shot to an instrumental velocity at 37½ feet of 920 plus or minus 20 feet per second, both within an average pressure limit of 11,000 pounds per square inch, will be supplied and assessed by the Sporting Arms and Ammunition Manufacturers' Institute. This ammunition will be used for qualifying the ranges and equipment and for making corrections in velocity and pressure firings in acceptance tests. Corrections in both velocity and pressure will be applied to the ammunition undergoing test, plus or minus as the case may be, based upon the deviation of the appropriate reference ammunition fired at time of test from the Sporting Arms and Ammunition Manufacturers' Institute assessed value of such reference ammunition in use at the time of signing of the ammunition purchase contract. Reference ammunition in the trap load shall be used for correcting all types and classes of ammunition except type III, class B, in which case, reference ammunition in the skeet load shall be used. Weapons will be disqualified from use for velocity and pressure measurements in acceptance tests if, when tested with reference ammunition, they deviate more than plus or minus 20 feet per second or plus or minus 200 pounds per square inch from the assessed velocity and pressure of such ammunition. Reference ammunition will be withdrawn from use in acceptance tests and replaced immediately by the Sporting Arms and Ammunition Manufacturers' Institute when, based upon the regular periodic, round robin firings, the average results are found to vary by more than plus or minus 20 feet per second from the assessed velocity or to exceed 11,000 pounds per square inch in pressure. The temperature of the ammunition when fired shall be 70° F. plus or minus 2° F.

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F–8d. Pressure shall be recorded with standard commercial lead cylinders, 0.325 inch diameter, 0.500 inch length (with the allowable tolerances), and the tarage table used shall be the appropriate tarage table, as shown in the Technical Committee Manual of the Sporting Arms and Ammunition Manufacturers' Institute.

F–8e. The center of the pressure piston hole shall be one inch forward from the face of the breech block. The diameter of the piston hole shall be 0.2250 with tolerance of plus 0.003 inch. The piston shall have a diameter of 0.2250 with tolerance of minus 0.0002 inch. Barrels shall be disqualified for further tests as soon as the "Not Go" plug gage, 0.2256 inch diameter, passes through the piston hole or the "Not Go" feeler gage, also 0.2256 inch diameter, enters the bottom of the piston hole for the height of the feeler, 0.123 inch.

F–8f. Pressure determinations of shot shell ammunition, except type I, shall be made either with or without the use of gas checks, at the option of the contracting officer. Gas checks shall be used in determining the pressure of type I.

F–8g. For all types of shot shells, other than type I, if gas checks be not used, the shell shall be prepared for pressure firings by cutting or piercing the paper under the piston hole. If gas checks be used, piercing is unnecessary. No paper stickers or grease shall be used for either method. If gas checks be not used, a full length piston, tangent with the wall of the chamber, shall be required. If gas checks be used, the piston shank shall be shortened to an amount equal to the outside height of the gas check.

F–8h. The metal case of type I shall be drilled under the piston hole and gas checks shall be used.

F–9. *Pattern.*—

F–9a. The firing for pattern with type I and type II and with type III, classes A and C, shall be made in a 30-inch standard barrel full choke shotgun, conforming to the specifications of the Sporting Arms and Ammunition Manufacturers' Institute. The pattern sheet shall be not less than 40 inches square and shall be located 120 feet plus or minus one foot in front of the muzzle. The diameter of the including circle for counting the shot holes shall be 30 inches. This test may be fired either independently of or simultaneously with the velocity and pressure test under paragraph F–8a unless it be found that the pattern paper interfere with the velocity measurements of the small sizes of shot.

F–9b. The firing for pattern with type III, class B, shall be made in an improved cylinder or "sheet" bored 26-inch standard barrel shotgun conforming to the specifications of the Sporting Arms and Ammunition Manufacturers' Institute. The pattern sheet shall be not less than 40 inches square and shall be located 75 feet plus or minus one foot in front of the muzzle. The diameter of the including circle for counting the shot holes shall be 30 inches. This test may be fired either independently of or simultaneously with the velocity and pressure test under paragraph F–8b unless it be found that the pattern paper interfere with the velocity measurements of the small sizes of shot.

F–9c. The temperature of the ammunition when fired shall be 70° F. plus or minus 2° F.

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F–10. *Function and casualty firing test.*—

F–10a. Shells, type I and type II, shall be tested in the shotguns named in paragraph F–5a; and the shotguns used shall conform to specification No. 52–18–1.

F–10b. Shells, type III, classes A, B, and C, shall be tested in the shotguns named in paragraph F–5b; and the guns used shall conform to specification No. 52–18–1.

F–10c. The shells for this test shall be shells which have not been used in the chambering test required under paragraph D–8h.

F–10d. The shells in this test shall be equally distributed in the guns required, placed in the magazines to capacity, chambered from the magazine by action of the guns, and fired. Failures to magazine and to chamber combined with the total defects discovered in the shells fired shall not exceed the percentages shown in table I.

F–11. *Defective shotgun.*—If any defect in firing be found to be caused by the shotgun used, such defect shall not be counted against the shells, but the shotgun shall be corrected or replaced and the test repeated. This applies also in any retest of shells.

F–12. *Retest.*

F–12a. Should the lot being tested fail on inspection or test, or both, upon the request of the contractor, approved by the contracting agency, the entire lot may be given a reinspection or retest, or both. The number of cartridges selected for such retests shall be twice the number of the original test.

F–12b. In case of failure of the sample of ammunition on one or more of the firing tests, the Government inspector shall limit the retest to the failing test or tests.

F–13. When shot shells are fired for velocity and pressure and with pattern in shotgun standard pressure barrels assembled to Universal receivers, shell casualties shall not be charged against the ammunition. When such defects occur during tests in shotgun standard pressure barrels, additional shells of the same type under test shall be substituted and the test completed with the required number of shells.

F–14. In the event of an appeal by the contractor against any decision of the Government inspector, the decision of the contracting agency shall be final.

G. PACKING AND MARKING FOR SHIPMENT

G–1. *Packing.*

G–1a. The shells shall be packed in accordance with the applicable drawings, unless commercial packing is specified in the invitation for bid, the contract, or purchase order.

G–1b. Shotgun shells, commercially packed, shall be in the usual commercial paper cartons which, in turn, shall be packed in the usual commercial wooden packing boxes; and both cartons and packing boxes shall be stamped "U. S. Property." When so required, the commercial wooden packing boxes shall be lined. Such lining shall be watertight and shall be approved by the contracting agency.

G–2. *Marking shipments.*—

G–2a. *Army.*—The packing boxes containing the shells shall be marked in accordance with applicable drawings.

G–2b. *Navy.*—The packing boxes containing the shells shall be marked in accordance with the drawings and in addition shall be

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marked in accordance with the requirements of the Navy Shipment Marking Handbook.

H. NOTES

H–1. The following shells are representative of the "low" metal heads (type III):

Federal Cartridge Co	"Monark."
Peters Cartridge Co	"Victor."
Remington Arms Co	"Shurshot."
Western Cartridge Co	"X Pert."
Winchester Repeating Arms Co	"Ranger."

H–2. Copies of specifications, Technical Committee Manual of the Sporting Arms and Manufacturers' Institute, and drawings should be obtained from the procuring agency or as directed by that agency.

H–3. Copies of the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Department of the Navy, Washington 25, D. C.

H–4. If the bidder desires to furnish a shell which in any way differs from any particular shell covered by this specification, a statement to that effect, describing fully the proposed difference, shall be submitted with his bid. In addition, he shall submit to the contracting agency full and sufficient evidence that such proposed shell is equal in all respects to the shell covered by this specification. If the contracting officer so directs, the bidder shall submit samples of such proposed shell for such tests as the contracting agency may require to assure himself further of their suitability.

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